

ABDELRAHMAN GAMAL ZAYED

AI / Machine Learning Intern | Artificial Intelligence Engineering Student

abdelrahman.gamal.zayed@gmail.com | +20 155 251 9691 | Ismailia, Egypt

LinkedIn: [linkedin.com/in/abdelrahman-gamal-zayed](https://www.linkedin.com/in/abdelrahman-gamal-zayed) | GitHub: github.com/Abdelra7manZayed | Portfolio:

abdelra7manzayed.github.io/Abdelrahman-portfolio/

SUMMARY

Artificial Intelligence Engineering student targeting AI / Machine Learning Intern roles with hands-on experience in supervised learning, deep learning, computer vision, and Natural Language Processing (NLP). Built data-driven and intelligent solutions using Python, Scikit-learn, TensorFlow, Keras, PyTorch basics, OpenCV basics, data collection, data cleaning, exploratory data analysis (EDA), feature extraction, data visualization, model evaluation, validation metrics, technical documentation, collaboration, motivation, and analytical thinking.

EDUCATION

Bachelor of Engineering in Artificial Intelligence Engineering | New Ismailia National University | Oct 2023 - Jul 2028 | Ismailia, Egypt

- GPA: 3.65 / 4.00

- Related fields: Artificial Intelligence, Machine Learning, Data Science, Computer Science, Deep Learning, Data Structures, and Object-Oriented Programming (OOP).

EXPERIENCE

Deep Learning Trainee | Egyptian Engineers Syndicate | Sep 2025 | Online

- Completed a 2-week training covering neural networks, validation metrics, overfitting control, transfer learning concepts, and deep learning applications.
- Documented model training logic, experiment results, and technical findings using analytical thinking.

Machine Learning Intern | National Telecommunication Institute (NTI) | Jul 2025 | Remote

- Built a diabetes prediction workflow using structured health data, data cleaning, preprocessing, feature engineering, model training, and evaluation.
- Evaluated 4 supervised learning algorithms - Logistic Regression, Random Forest, Support Vector Machine (SVM), and Decision Tree - using accuracy, precision, recall, F1 score, and confusion matrix validation metrics.

AI & Machine Learning Trainee | SprintUp Program by Sprints | Apr 2025 - Jun 2025 | Remote

- Developed an end-to-end supervised learning workflow using Python, Scikit-learn, exploratory data analysis, data visualization, and result interpretation.
- Transformed raw data into model-ready features and documented findings for collaborative AI project development.

RESEARCH EXPERIENCE

Research Paper - Hybrid Metaheuristic Model for High-Fidelity Solar Power Forecasting | Under Review

- Co-authored an under-review paper integrating Extreme Learning Machine (ELM) with Improved Harris Hawks Optimization (IHO) for solar power forecasting.
- Benchmarked CatBoost, XGBoost, Transformer, and standalone ELM using R2, Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

PROJECTS

Diabetes Prediction Classification Model | Machine Learning | Healthcare Data

- Built diabetes risk classification models using preprocessing, feature selection concepts, hyperparameter tuning basics, cross-validation basics, and model comparison.
- Reported performance using 5 evaluation metrics: accuracy, precision, recall, F1 score, and confusion matrix; focused on recall and class-level error analysis.

Multilingual Fake News Classification | Natural Language Processing (NLP) | Arabic & English

- Built a multilingual fake news text classification workflow for Arabic and English using data cleaning, text preprocessing, feature extraction, EDA, and model evaluation.
- Documented technical findings and classification results to support reproducible Natural Language Processing (NLP) project development.

Facial Emotion Recognition Model | Deep Learning | Computer Vision

- Developed a 7-class facial emotion image classification workflow using TensorFlow/Keras, Convolutional Neural Network (CNN) concepts, data augmentation, validation monitoring, and transfer learning concepts.
- Applied class-level error analysis, confusion matrices, overfitting checks, and OpenCV basics to improve computer vision model evaluation quality.

SKILLS

Programming & Tools: Python, C, C++, Jupyter Notebook, Git, GitHub, PyCharm, technical documentation.

Machine Learning & Data Science: Machine Learning, supervised learning, classification, regression basics, data collection, data cleaning, exploratory data analysis (EDA), data visualization, feature engineering, feature selection, hyperparameter tuning, cross-validation, model evaluation, validation metrics.

Libraries & Models: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, TensorFlow, Keras, PyTorch basics, OpenCV basics, Logistic Regression, Random Forest, Support Vector Machine (SVM), Decision Tree, XGBoost basics.

AI Areas: Deep Learning, Neural Networks, Computer Vision, image classification, Natural Language Processing (NLP), text preprocessing, feature extraction, text classification, model deployment basics, intelligent solutions, data-driven solutions.

Soft Skills: Problem-solving, analytical thinking, collaboration, teamwork, motivation, time management, fast learning, communication. Languages: Arabic native; English intermediate.

LEADERSHIP & VOLUNTEERING

AI Head | Microsoft Students Club - Suez Canal University (SCU) | Present

- Led AI-focused student activities and peer learning in machine learning, deep learning, Natural Language Processing (NLP), and computer vision.

Volunteer | Egyptian Youth Entity (EYE) | Present

- Supported youth and community development initiatives through teamwork, collaboration, communication, and organized volunteer activities.